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Project 1 report

I Introduction

The objective of this project is to design and evaluate a recommendation system based solely on textual reviews from TripAdvisor. Given one place (hotel, restaurant, attraction, or attraction product), the system must retrieve the most similar place using only user-generated reviews, without relying on metadata during the modeling phase.

II DATA CLEANING

The original dataset contained reviews written in multiple languages. Since textual similarity methods are highly sensitive to vocabulary and linguistic structure, mixing languages would negatively affect model performance.

Therefore, **only English reviews were retained**. This filtering step ensured linguistic consistency and reduced noise introduced by multilingual content.

Although this reduced the total number of reviews, the remaining dataset was sufficiently large to support reliable experimentation.

```
print(reviews.shape)
print(places.shape)
```

```
(340385, 21)
(3761, 60)
```

```
reviews_en = reviews[reviews["langue"] == "en"].copy()

print(reviews_en.shape)
```

```
(153071, 21)
```

After english language restriction, **we keep around 45% of all the reviews that is enough**.

Reviews with missing text or empty content **were removed**. Only valid textual reviews were kept for document construction.

For each place:

- Reviews were grouped using the place identifier.
- Selected reviews were concatenated into a single textual document.
- The resulting document represents the textual description of the place.

After constructing the place-level documents, the dataset was split into two subsets:

- 50% of places used as queries
- 50% used as candidate recommendations

III Model + evaluation

The ranking evaluation is computed by measuring, for each query place, the position of the first correctly matching candidate in the similarity-based ranking, and then averaging this position across all queries to obtain the mean ranking error.

```
print(errors)
```

```
[0, 0, 0, 0, 0, 0, 3, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 4, 0, 0, 0, 2, 0, 0, 1, 0, 0, 0, 0, 0, 3, 1, 1, 0, 2, 0, 0, 0, 2, 0, 0, 0, 0, 0, 1, 0,
3, 10, 0, 0, 0, 1, 30, 0, 0, 0, 0, 0, 0, 0, 3, 0, 0, 27, 0, 1, 2, 0, 1, 0, 0, 0, 0, 0, 9, 0, 0, 0, 0, 0, 1, 0, 0, 1, 1, 0, 0, 0, 1, 1, 1, 0, 3, 0, 0,
0, 2, 1, 0, 0, 0, 1, 0, 1, 15, 0, 2, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 2, 0, 0, 2, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 1, 3, 4, 0, 0, 0,
0, 0, 0, 0, 1, 3, 0, 1, 0, 0, 0, 0, 9, 0, 1, 0, 0, 2, 0, 0, 0, 0, 0, 1, 0, 0, 3, 0, 1, 0, 7, 0, 0, 0, 0, 1, 0, 0, 0, 2, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
1, 0, 2, 12, 0, 0, 0, 1, 0, 0, 0, 0, 4, 0, 0, 0, 0, 1, 0, 2, 0, 0, 0, 0, 1, 0, 0, 1, 0, 1, 0, 1, 0, 5, 0, 0, 0, 0, 1, 0, 2, 1, 0, 0, 0, 0, 0, 4, 1, 0,
0, 0, 0, 2, 2, 5, 1, 15, 0, 1, 21, 0, 1, 0, 0, 0, 11, 1, 1, 1, 0, 0, 1, 0, 0, 2, 0, 0, 0, 1, 0, 3, 0, 8, 1, 4, 0, 0, 0, 17, 33, 0, 1, 0, 0, 0, 0, 3, 1,
1, 0, 0, 0, 0, 0, 2, 0, 0, 2, 0, 1, 4, 0, 0, 2, 4, 0, 0, 0, 0, 0, 0, 0, 6, 0, 0, 0, 0, 1, 1, 0, 0, 0, 4, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 4,
0, 0, 0, 0, 0, 7, 1, 8, 5, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 2, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 0, 1, 0, 0, 0, 0, 13, 1, 0, 0, 1,
0, 0, 0, 1, 0, 0, 0, 0, 0, 2, 0, 4, 0, 0, 1, 4, 0, 2, 0, 1, 1, 0, 1, 0, 0, 3, 0, 5, 0, 2, 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 1,
0, 0, 0, 0, 1, 1, 1, 1, 0, 0, 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 8, 0, 0, 0, 5, 3, 0, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 1, 1, 0, 0, 0, 0, 7, 0, 0, 1, 0, 0, 0, 0, 0, 1, 2, 0, 0, 0, 5, 1, 0, 0, 1, 0, 0, 0, 0, 1, 0, 0, 0, 0, 3, 0, 0, 1, 23, 2, 0, 0, 0, 0, 0, 0, 0, 0, 2,
0, 0, 4, 2, 4, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 2, 0, 1, 7, 1, 0, 0, 3, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0, 3, 2, 0, 0, 0,
0, 0, 0, 0, 4, 0, 0, 0, 0, 1, 1, 2, 0, 0, 2, 0, 0, 0, 0, 0, 2, 1, 0, 2, 0, 0, 3, 0, 0, 0, 0, 0, 0, 0, 1, 0, 7, 0, 0, 0, 20, 0, 6, 0, 0, 0, 0, 0, 0, 0,
0, 5, 0, 0, 1, 1, 0, 0, 1, 0, 0, 0, 4, 1, 0, 0, 1, 0, 0, 2, 1, 7, 0, 0, 0, 2, 0, 0, 0, 0, 0, 0, 1, 2, 0, 0, 0, 1, 2, 1, 1, 0, 0, 0, 2, 0, 0, 0, 0, 0,
0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 2, 0, 0, 0, 1, 0, 0, 0, 3, 2, 0, 0, 0, 25, 1, 1, 0, 0, 9, 1, 0, 0, 0, 0, 1, 1, 1, 0, 0, 3, 0, 5, 0, 0, 3, 3, 0, 1,
1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 1, 0, 3, 0, 0, 4, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 2, 0, 0, 0, 0, 22, 0, 0, 4, 0, 0, 0, 0, 14, 0,
0, 0, 4, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 1, 0, 0, 0, 3, 0, 4, 2, 0, 1, 0, 0, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0,
0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 0, 1, 5, 1, 0, 0, 1, 1, 0, 0, 2, 0, 2, 0, 7, 0, 0, 1, 0, 0, 0, 0, 2, 0, 0, 5, 1, 0, 4, 0, 1, 0,
0, 3, 0, 0, 0, 1, 0, 0, 0, 0, 3, 0, 0, 0, 0, 3, 0, 2, 0]
```

For exemple we take the average number of this kind of list. (length = 1835 because 1835 differents places)

- For the first level we are only matching the Type of place (H,A,AP,R)
- For the second level we are matching on the most interesting metadata choose by myself and seems to be the best matching indicators that a client should take into account :
  - Hotel : price range
  - Restaurant : restaurant_type, restaurant_type_cuisine
  - Attraction : activiteSubCategorie, activiteSubType
  - Attrction Product : ap_primary_supplier_subtype

a) BM25 Baseline

As a baseline, we implemented the BM25 ranking function, a classical probabilistic information retrieval model. BM25 scores documents based on term frequency, inverse document frequency, and document length normalization.

Each place was represented by the concatenation of all its reviews. Similarity between a query place and candidate places was computed using the BM25 scoring function.

Level 1 Error: 0.94
Level 2 Error: 5.67

Analysis:

BM25 performs reasonably well for coarse-grained recommendations (Level 1), successfully distinguishing between major place types such as hotels and restaurants. However, performance significantly degrades at Level 2. This indicates that while lexical overlap is sufficient for broad categorization, it is

less effective for fine-grained distinctions such as cuisine types or attraction subcategories.

b) TF/IDF

A TF-IDF vectorizer using unigrams was then applied to represent each place as a weighted sparse vector. Cosine similarity was used for ranking.

Level 1 Error: 0.89

Level 2 Error: 3.77

Analysis:

Compared to BM25, TF-IDF improves fine-grained performance considerably. The weighting scheme better highlights discriminative terms while reducing the impact of common words.
However, unigram representation still fails to capture multi-word expressions (e.g., “fine dining”, “theme park”), limiting its ability to distinguish subtle differences between places

c) Improved TF-IDF (Unigrams + Bigrams)

To enhance textual representation, bigrams were introduced and document frequency filtering was applied (min_df and max_df constraints). This allows the model to capture meaningful multi-word expressions and reduce noise from overly frequent or extremely rare terms.

Level 2 Error: 3.38

Analysis:

Adding bigrams improves performance, particularly at Level 2. This confirms that fine-grained distinctions often rely on multi-word expressions rather than isolated terms. The inclusion of frequency thresholds further stabilizes the vocabulary and reduces noise.

d) Sentence Embeddings

To explore semantic similarity beyond lexical overlap, transformer-based sentence embeddings were tested. Each place document was encoded into a dense vector representation, and cosine similarity was used for ranking.

Level 2 Error: 4.24

Analysis:

Contrary to expectations, embeddings performed worse than TF-IDF. This suggests that in this specific task, lexical patterns are more informative than deeper semantic similarity.

e) TF-IDF with Balanced Review Representation (10 Reviews)

The maximum number of reviews per place was further reduced to 10 to test whether a more concise representation improves discrimination.

Level 2 Error: 2.13

Analysis:

This configuration achieved the best overall performance. Reducing the number of reviews further enhanced signal clarity and minimized generic or repetitive content.
This experiment demonstrates that excessive textual information may introduce noise rather than improve recommendation quality.

IV CONCLUSION

BEST MODEL

	Model	Data Strategy	Level 1 Error	Level 2 Error	Notes
5	TF-IDF (1,2-grams improved)	Max 10 reviews per place	nan	2.132000	Best performing configuration
4	TF-IDF (1,2-grams improved)	Max 20 reviews per place	nan	2.487300	Balanced document length
2	TF-IDF (1,2-grams improved)	All reviews	nan	3.375100	Bigrams + frequency filtering
1	TF-IDF (unigrams)	All reviews	0.889900	3.773400	Basic TF-IDF representation
3	Sentence Embeddings	All reviews	nan	4.240900	Transformer-based semantic embeddings
0	BM25	All reviews	0.938900	5.666700	Probabilistic IR baseline